

Instructions: Answer all questions. Show your work for each problem by following the four problem-solving steps.

Kinematics Formulas

$$s_{avg} = \frac{d_{total}}{t_{total}} \quad v_{avg} = \frac{\Delta x_{total}}{t_{total}} \quad a = \frac{\Delta v}{t} = \frac{v - v_0}{t}$$
$$v = v_0 + at \quad t = \frac{v - v_0}{a}$$

Reminders

- Average speed uses total **distance**; average velocity uses total **displacement**.
- Add displacements in the same direction; subtract displacements in opposite directions.
- Include the direction (or a $+/-$ sign) for velocity and acceleration.
- On a position–time graph, the slope is the velocity.
- A steeper slope means a greater speed; a flat line means zero velocity.

Problem-Solving Steps

1. What am I given? What am I finding?
2. What equation connects them?
3. Plug in and solve.
4. Check the units and direction (if applicable).

1. A runner jogs 400 m in 100 s, then walks 200 m in 50 s. What is her average speed for the whole trip?
2. A delivery van drives 240 m east in 20 s, then 160 m east in 20 s. What is its average velocity for the whole trip?
3. A hiker walks 400 m east in 320 s, then turns around and walks 100 m west in 90 s. What is her average velocity for the whole trip?

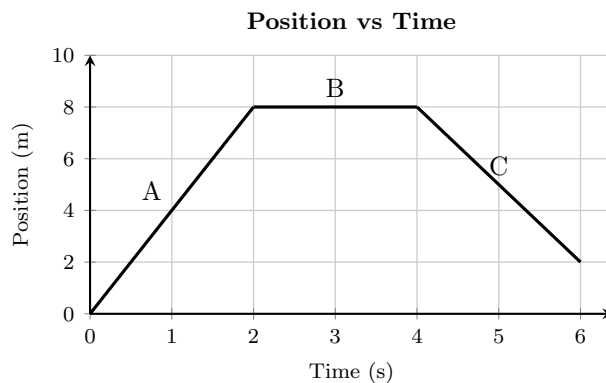
4. A car speeds up from 10 m/s to 30 m/s in 5 s. What is its acceleration?

5. A car starts from rest and accelerates at 2 m/s^2 for 8 s. What is its final velocity?

6. A car speeds up from 5 m/s to 25 m/s with a constant acceleration of 4 m/s^2 . How long does it take to reach 25 m/s?

7. A car drives around a circular track at a constant speed of 15 m/s. Is its velocity constant? Explain.

8. The position–time graph below shows the motion of a remote-controlled car along a straight track. The motion has three labeled segments, A, B, and C.

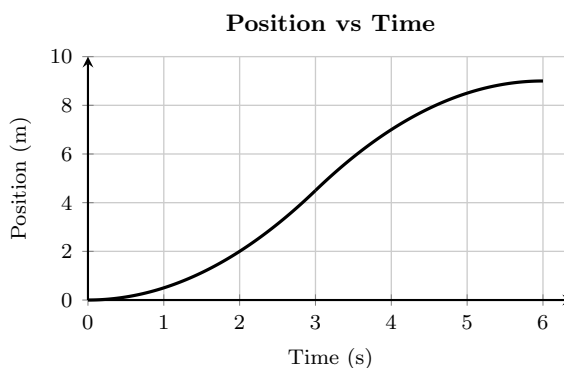


- (a) For each segment, state whether the velocity is positive, negative, or zero.

A: _____ B: _____ C: _____

- (b) Explain how the slope of each segment tells you the sign of the velocity.

9. The position–time graph below shows an object moving along a straight line. The curve bends upward from $t = 0$ to $t = 3$ s and bends downward from $t = 3$ s to $t = 6$ s.



- (a) During which time interval is the object speeding up?

- (b) During which time interval is the object slowing down?

- (c) Explain how the changing slope of the curve tells you whether the object is speeding up or slowing down.